IKBKE fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes IKBKE gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 3/6 fusions with known frame status (50.0%) were in-frame. 4/11 fusions (36.4%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.0666667, not statistically significant at alpha=0.05). No literature benchmark is configured for IKBKE.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was NAV1, observed in 2 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, p=0.0666667 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.564356.

Domain disruption

Domain-disruption analysis was skipped: IKBKE has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

20% of 5 fusions with this gene as the 5' partner join at exon 12, against 2% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.742574, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 10 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 359 aa (exon 10; label-permutation p=0.531469). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 115 aa from the nearest configured domain boundary at 244 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=3) and "not_retained" (n=7). For Frame_status, retained: 2/3 (66.7%, 95% CI 20.8%-93.9%); not_retained: 0/2 (0.0%,

95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 19.3333 vs 19.4286, p=0.993168 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for IKBKE in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: IKBKE has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: IKBKE has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (6 row(s) in window_scan, 3 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NAV1	2	0.16666666666666666	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.13351961785262864	1
IL20	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	2
NFASC	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	3
RABGAP1L	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	4
RASAL2	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	5
RASSF5	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	6
RBBP5	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	7
SMYD2	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	8
SRGAP2	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	9
TCEA3	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	10
THBS3	1	0.08333333333333333	0.024844651811015122	None	None	0.3057658865145485	confidence_certainty, domain_retention, recurrence	0.09505807939109019	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
76	0	3	1	6	0.0	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
120	0	3	2	5	0.0	1.0	-0.0
206	0	3	3	4	0.0	0.475	0.3233063903751334
359	0	3	4	3	0.0	0.19999999999999998	0.6989700043360189
395	1	2	4	3	0.375	1.0	-0.0
447	2	1	4	3	1.5	1.0	-0.0
539	2	1	5	2	0.8	1.0	-0.0
572	2	1	6	1	0.3333333333333333	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

2	1
0	7

frame_domain_contingency_tables_by_tumor_type

stratum	table
GBC	[0, 1], [0, 0]
HGSOC	[0, 0], [0, 2]
IDC	[0, 0], [0, 3]
ILC	[1, 0], [0, 0]
MEL	[1, 0], [0, 0]
PRAD	[0, 0], [0, 1]
SKCM	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	10	3	3	4	0.13191489361702127

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
3	5	11	0.45454545454545453

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
IL20	1	1	1	1
NAV1	2	2	2	1
NFASC	1	1	1	1
RABGAP1L	1	1	1	1
RASAL2	1	1	1	1
RASSF5	1	1	1	1
RBBP5	1	1	1	1
SMYD2	1	1	1	1
SRGAP2	1	1	1	1
TCEA3	1	1	1	1
THBS3	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
1	206648000	206649000	2	EVT-P-0052992-T01-IM6-18, EVT-P-0058226-T01-IM6-21
1	206652000	206653000	2	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24
1	206650000	206651000	1	EVT-P-0054307-T01-IM6-28
1	206653000	206654000	1	EVT-P-0051760-T01-IM6-15
1	206661000	206662000	1	EVT-P-0067348-T01-IM7-25
1	206664000	206665000	1	EVT-P-0022004-T02-IM6-27
1	206666000	206667000	1	EVT-P-0053124-T01-IM6-17
1	206667000	206668000	1	EVT-P-0027193-T03-IM6-14

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
682	2	2	361	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	4	1	5	0.2	0.020202020202020204	1.013181238358318
five_prime	5	1	5	0.2	0.04978625203344304	0.6471459539388933
five_prime	10	1	5	0.2	0.035069799114742936	0.7865515826533809
five_prime	12	1	5	0.2	0.016078386864903718	1.108752423443961
five_prime	20	1	5	0.2	0.0247418000226989	0.929086932619802
three_prime	7	1	5	0.2	0.018007793288692165	1.0612100149207309
three_prime	10	1	5	0.2	0.030038209813490712	0.8494428124491662
three_prime	16	1	5	0.2	0.10153974198918019	0.3824295025646541
three_prime	17	1	5	0.2	0.10834941172019824	0.3601185097360637
three_prime	21	1	5	0.2	0.02319070858396701	0.9558573399988561

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
339	539	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
359	559	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
372	572	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
395	595	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
482	682	200	4	6	1	2	0.6666666666666666	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25
539	739	200	4	6	1	2	0.6666666666666666	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
339	539	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
372	572	200	4	6	2	1	5.0	0.5	0.3010299956639812	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0067348-T01-IM7-25
482	682	200	4	6	1	2	0.6666666666666666	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027193-T03-IM6-14	P-0027193-T03-IM6	P-0027193	Prostate Cancer	PRAD	IKBKE-IL20	IL20	unknown	three_prime	206667107	21	682	True	682	True	exact match	lost	0.0	False		Protein kinase domain		IL20 (NM_018724) - IKBKE (NM_014002) Rearrangement : c.398:IL20_c.2046-146:IKBKE | Protein Fusion: mid-exon {IL20:IKBKE}	NA
EVT-P-0051760-T01-IM6-15	P-0051760-T01-IM6	P-0051760	Melanoma	MEL	IKBKE-NAV1	NAV1	in-frame	five_prime	206653556	12	447	True	446	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+100:IKBKE_c.1226+27872:NAV1dup | Protein Fusion: in frame {IKBKE:NAV1}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051760-T02-IM6-16	P-0051760-T02-IM6	P-0051760	Melanoma	SKCM	IKBKE-NAV1	NAV1	in-frame	five_prime	206653558	12	447	True	446	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+102:IKBKE_c.1226+27873:NAV1dup | Protein Fusion: in frame {IKBKE:NAV1}	NA
EVT-P-0053124-T01-IM6-17	P-0053124-T01-IM6	P-0053124	Hepatobiliary Cancer	GBC	IKBKE-RABGAP1L	RABGAP1L	out-of-frame	five_prime	206666746	20	682	True	681	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - RABGAP1L (NM_014857) rearrangement: c.2045+35:IKBKE_c.1710+64980:RABGAP1Ldup | Protein Fusion: out of frame {IKBKE:RABGAP1L}	NA

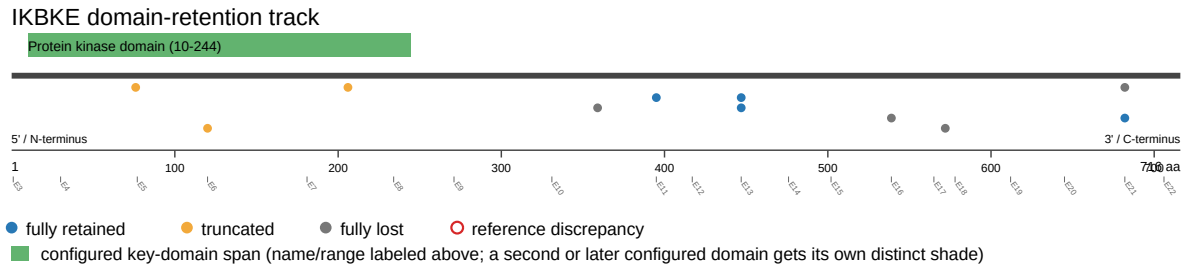
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052992-T01-IM6-18	P-0052992-T01-IM6	P-0052992	Melanoma	SKCM	IKBKE-RBBP5	RBBP5	out-of-frame	five_prime	206648412	5	120	True	119	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.472340425531 9149	True			Protein kinase domain	IKBKE (NM_014002) - RBBP5 (NM_005057) rearrangement: c.358+75:IKBKE_c.218+2153:RBBP5inv | Protein Fusion: out of frame {IKBKE:RBBP5}	NA
EVT-P-0058226-T01-IM6-21	P-0058226-T01-IM6	P-0058226	Breast Cancer	IDC	IKBKE-THBS3	THBS3	unknown	five_prime	206648064	4	76	True	76	True	exact match	disrupted	0.285106382978 7234	True			Protein kinase domain	IKBKE (NM_014002) - THBS3 (NM_007112) rearrangement: c.229-144:IKBKE_c.2579:THBS3inv | Protein Fusion: mid-exon {IKBKE:THBS3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001902-T01-IM3-23	P-0001902-T01-IM3	P-0001902	Breast Cancer	ILC	NFASC-IKBKE	NFASC	in-frame	five_prime	206652479	10	395	True	394	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NFASC (NM_001005388) Duplication : c.1183+3_c.2782+2783dup | Protein fusion: in frame (IKBKE-NFASC)	NA
EVT-P-0019364-T01-IM6-24	P-0019364-T01-IM6	P-0019364	Breast Cancer	IDC	RASAL2-IKBKE	RASAL2	unknown	three_prime	206652369	10	359	False	359	True	exact match	lost	0.0	False		Protein kinase domain		RASAL2 (NM_170692) - IKBKE(NM_014002) Rearrangement : c.1628-181:RASAL2_c.1076:IKBKEdel | Protein Fusion: mid-exon {RASAL2:IKBKE}	NA

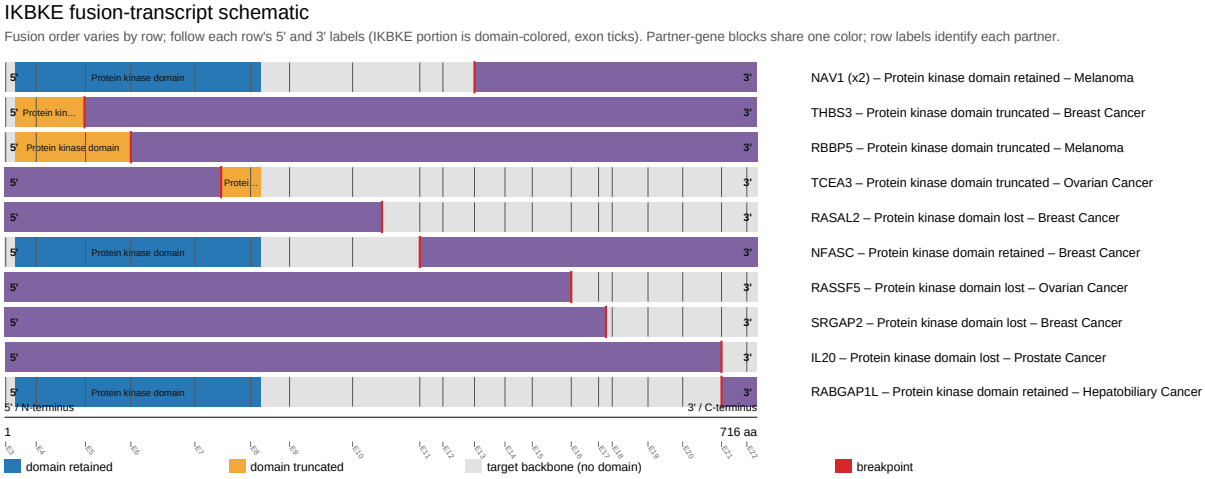
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067348-T01-IM7-25	P-0067348-T01-IM7	P-0067348	Ovarian Cancer	HGSOC	RASSF5-IKBKE	RASSF5	out-of-frame	three_prime	206661210	16	539	True	539	True	exact match	lost	0.0	False		Protein kinase domain		RASSF5 (NM_182663) - IKBKE (NM_014002) rearrangement: c.579+1935:RASSF5_c.1617-41:IKBKEdup | Protein Fusion: out of frame {RASSF5:IKBKE}	NA
EVT-P-0022004-T02-IM6-27	P-0022004-T02-IM6	P-0022004	Breast Cancer	IDC	SRGAP2-IKBKE	SRGAP2	unknown	three_prime	206664172	17	572	False	572	True	exact match	lost	0.0	False		Protein kinase domain		SRGAP2 (NM_015326) - IKBKE (NM_014002) rearrangement: c.2085:SRGAP2_c.1714:IKBKEdel | Protein Fusion: mid-exon {SRGAP2:IKBKE}	NA
EVT-P-0054307-T01-IM6-28	P-0054307-T01-IM6	P-0054307	Ovarian Cancer	HGSOC	TCEA3-IKBKE	TCEA3	unknown	three_prime	206650096	7	206	False	206	True	exact match	disrupted	0.165957446808 51063	True			Protein kinase domain	TCEA3 (NM_003196) - IKBKE (NM_014002) rearrangement: c.69+346:TCEA3_c.616:IKBKEinv | Protein Fusion: mid-exon {TCEA3:IKBKE}	NA

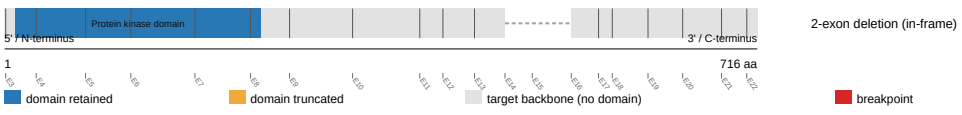
Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.

intragenic deletion schematic